Week 47: Gentle-step landscapes

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-47-facilitator.pdf, whose source is editable/facilitator-src/content.json with route-note.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the envelope proof are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Different landscapes between two clues of height 1	P1	P1	P1	first
How high and how low can the starred marker go?	P2	P2	P2	first
A right clue that forces every marker	P3	P3	P3	first
Which clue sets can be completed?	P4	P4	P4	return
Lowest and highest landscapes on seven sites	P5	P5	P5	return
Fewest one-level moves between two landscapes	P6			return
Invent two clues for a friend	P7			return
Two clues that force one landscape, and two that do not		P6		return
Possible heights for a clue four steps away		P7		return
Allowed heights around one clue and around two			P6	return
Must pairwise-consistent clues be fillable?			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P7; Grades 2–3 P4–P7; Grades 4–5 P4–P7.

Key, one entry per problem

Different landscapes between two clues of height 1

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. With clues 1 at sites 0 and 4, many landscapes fit, and the middle site, two steps from each clue, can be any height from 0 to 3.

How high and how low can the starred marker go?

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. With clues 0 at site 0 and 2 at site 4, the middle site can be exactly 0, 1 or 2: the left clue caps it at 2 and heights cannot go below 0.

A right clue that forces every marker

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. With clues 0 and 4 four steps apart, every step must rise by 1, so the ramp 0, 1, 2, 3, 4 is forced; 4 is the only right-clue height that forces every site.

Which clue sets can be completed?

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. Clues can be completed exactly when every pair differs by at most their distance: 0 and 3 two steps apart cannot, while 3 and 0 four steps apart can.

Lowest and highest landscapes on seven sites

K–1 P5 · Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. With clues 1, 3, 1 at sites 0, 4, 6 there are 10 landscapes; the pointwise lowest (1,0,1,2,3,2,1) and highest (1,2,3,4,3,2,1) are themselves landscapes, with the least and greatest totals 10 and 16.

Fewest one-level moves between two landscapes

K–1 P6 · return route

Idea. Moving one marker one level at a time, the fewest moves from 1,2,1,2,1 to 1,0,0,0,1 is the total change, 5, though lowering the middle first is illegal.

Invent two clues for a friend

K–1 P7 · return route

Idea. Two clues can be completed when their heights differ by at most their distance; on this five-site board only end clues 0 and 4, in either order, force a single landscape.

Two clues that force one landscape, and two that do not

Grades 2–3 P6 · return route

Idea. On sites 0 to 6 with no ceiling, two clues force a whole landscape only when they are the end sites and differ by 6: h(0)=0, h(6)=6 or its mirror.

Possible heights for a clue four steps away

Grades 2–3 P7 · return route

Idea. A clue d steps from a clue of height a can have any height from max(0, a - d) to a + d; four steps from height 1 that is 0 to 5.

Allowed heights around one clue and around two

Grades 4–5 P6 · return route

Idea. A single clue of height a allows every height from max(0, a - d) to a + d at distance d; with several clues a site's range is the overlap of these, and heights allowed at each site separately need not fit together.

Must pairwise-consistent clues be fillable?

Grades 4–5 P7 · return route

Idea. Yes: if every pair of clues differs by at most their distance, the lower envelope $L(i) = \max(0, \max_c(a_c - |i - c|))$ is a landscape through every clue (discrete McShane extension).